

2.4.37

EE25BTECH11041 - Naman Kumar

Question:

Find a vector of magnitude 6, which is perpendicular to both the vectors $2\hat{i} - \hat{j} + 2\hat{k}$ and $4\hat{i} - \hat{j} + 3\hat{k}$

Solution:

Given Vectors

$$\mathbf{A} = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}, \mathbf{B} = \begin{pmatrix} 4 \\ -1 \\ 3 \end{pmatrix} \quad (1)$$

Let required vector be,

$$\mathbf{C} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \quad (2)$$

Using Inner Product,

$$\mathbf{C}^T \cdot \mathbf{A} = 0 \text{ and } \mathbf{C}^T \cdot \mathbf{B} = 0 \quad (3)$$

$$\mathbf{C}^T \cdot \mathbf{A} = 2x - y + 2z = 0 \quad (4)$$

$$\mathbf{C}^T \cdot \mathbf{B} = 4x - y + 3z = 0 \quad (5)$$

Using Rank to Analyze the system

$$2x - y + 2z, 4x - y + 3z \quad (6)$$

$$\mathbf{A} = \begin{pmatrix} 2 & -1 & 2 \\ 4 & -1 & 3 \end{pmatrix} \quad (7)$$

Using Row Transformation $R_2 \rightarrow R_2 - 2R_1$

$$\mathbf{A} = \begin{pmatrix} 2 & -1 & 2 \\ 0 & 1 & -1 \end{pmatrix} \quad (8)$$

Since rank of matrix is $2(\leq 3)$, there are infinite many solutions $R^3 \rightarrow R^2$

Solving equations to get general vectors

From the row echelon form, we can write the new system of equation:

$$2x - y + 2z = 0 \quad (9)$$

$$y - z = 0 \quad (10)$$

Substitute $y=z$ in (9)

$$2x - z + 2z = 0 \quad (11)$$

$$z = -2x \quad (12)$$

Therefore vector $\mathbf{C}$ is

$$\mathbf{C} = (x, -2x, -2x) = x(1, -2, -2) \quad (13)$$

Now getting vector with magnitude 6

$$|C| = 6 \quad (14)$$

$$|x(1, -2, -2)| = 6 \quad (15)$$

$$|x| \sqrt{(1)^2 + (-2)^2 + (-2)^2} = 6 \quad (16)$$

$$|x| \sqrt{1 + 4 + 4} = 6 \quad (17)$$

$$|x| \sqrt{9} = 6 \quad (18)$$

$$|x| = 2 \quad (19)$$

Therefore final vectors are

$$C_1 = \begin{pmatrix} -2 \\ 4 \\ 4 \end{pmatrix}, C_2 = \begin{pmatrix} 2 \\ -4 \\ -4 \end{pmatrix} \quad (20)$$

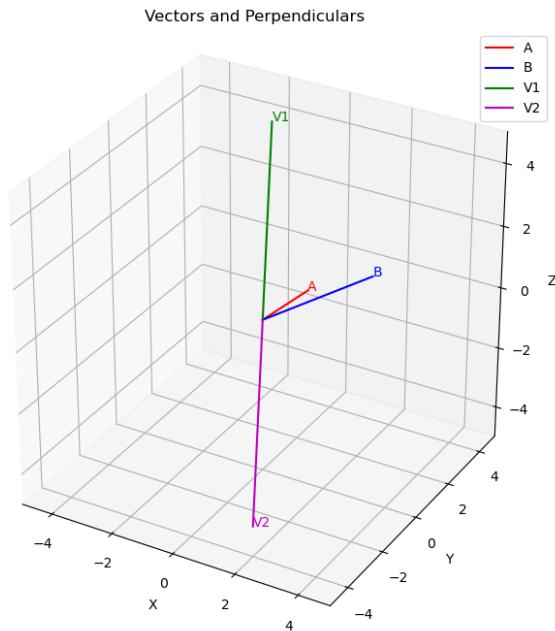


Fig. 1